

Processing Audio Live on a Raspberry Pi

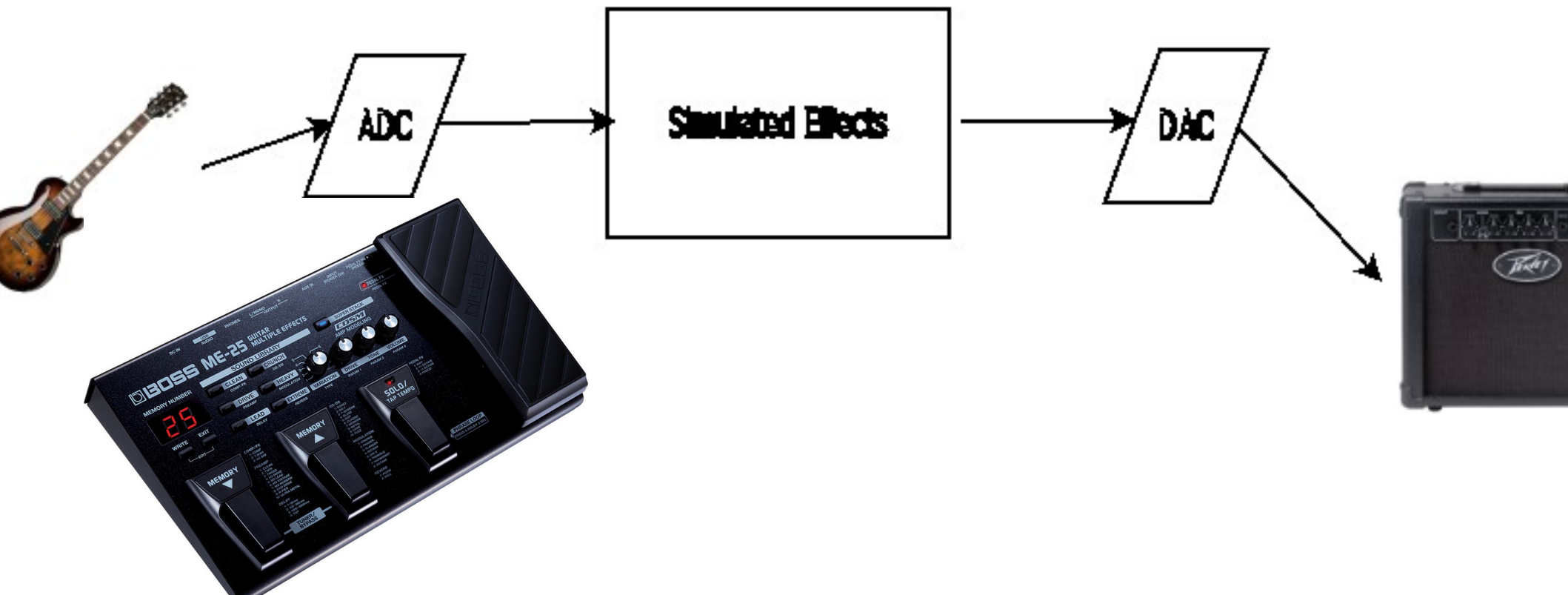


Electric Guitar

- Analogue electrical signal
- Effects “pedals”
- Amplification



Digital Effects



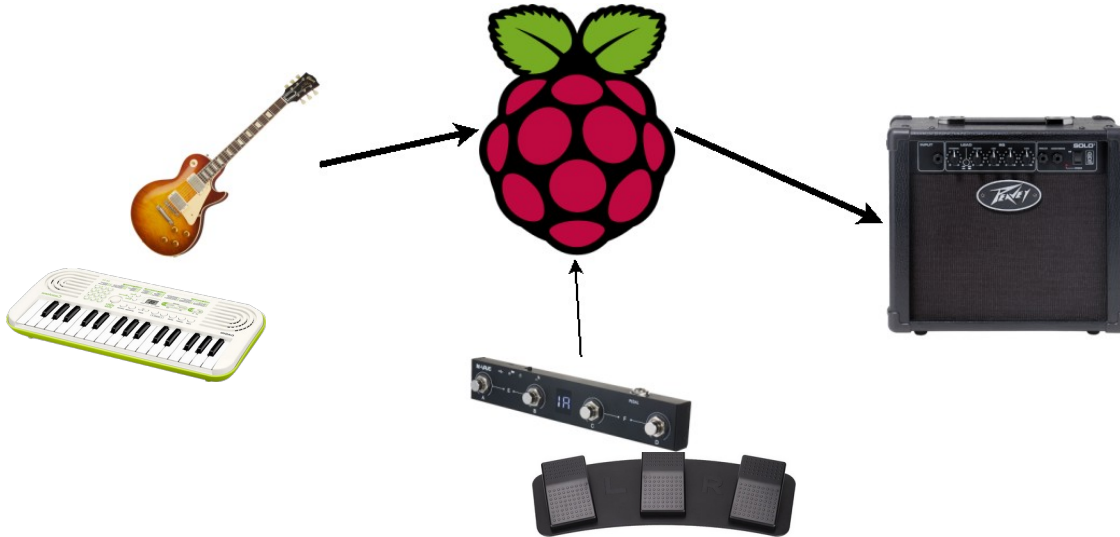
Desire

- Audio I/O
 - High quality
 - Low latency
 - Processing capacity of Pi 5 not a problem, yet
- Switch < 100ms
 - Audible perception



System Design(!)

- Electric guitar as source
 - Or any other physical instrument producing analogue electrical signals
 - E.g: vocals, theremins, “vintage” electronic instruments



Challenges

- **Sound**

- On board sound processing poor quality up to Raspberry Pi 5
 - Output only – need input too
- Raspberry Pi 5 dispenses with any audio interface

- **Pedal**

- Pedal must be robust – beer spills, heavy feet
- Pedal must be available/reasonable price – not going to build one myself

Mark 1

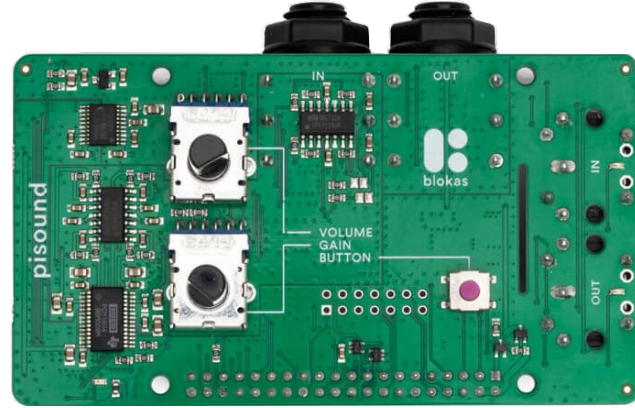
- **PI Sound – Hat for Raspberry Pi**
 - “ultra-low latency high-quality sound card...
 - ...and MIDI interface
- **USB Keyboard Pedal**
 - Cheap



Problems

- **Pi Sound is delectate**

- Stereo I/O – odd decision
- MIDI DIN plugs unnecessary



USB Foot Keyboard is obviously unsuited



Important Digression

- **Modep: mod-ui**
 - LV2 simulators
 - Jack Audio pipes
- Some other stuff



Digression ii: Modep/Modhost

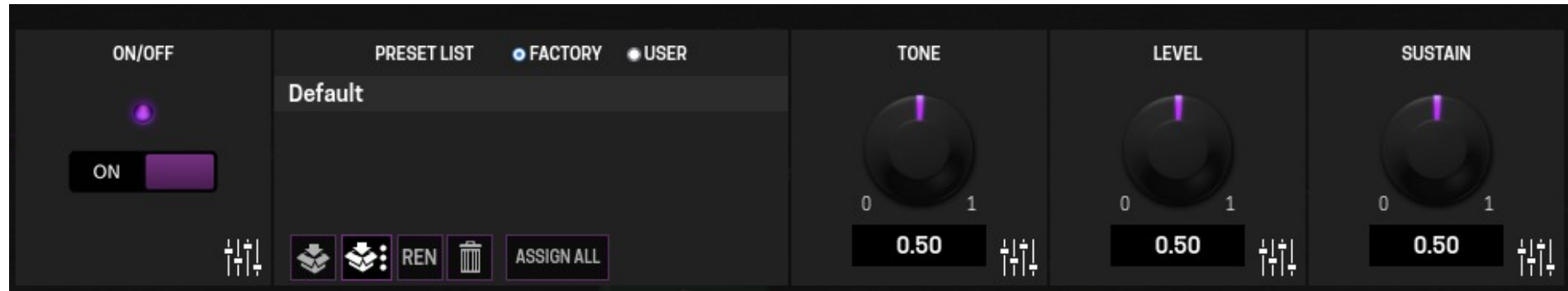
LV2 Simulators

Controlled by a host

Control host with Unix socket

Each simulator has a numeric identifier 1,2,3...

```
add add http://moddevices.com/plugins/mod-devel/BigMuffPi 7
param_set 7 Tone 0.500000
param_set 7 Level 0.500000
param_set 7 :bypass 0
param_set 7 Sustain 0.500000
```



Solutions: Sound



Standard case

USB Sound card



Cheap and cheerful



Not cheap – low latency

Solutions: Pedal

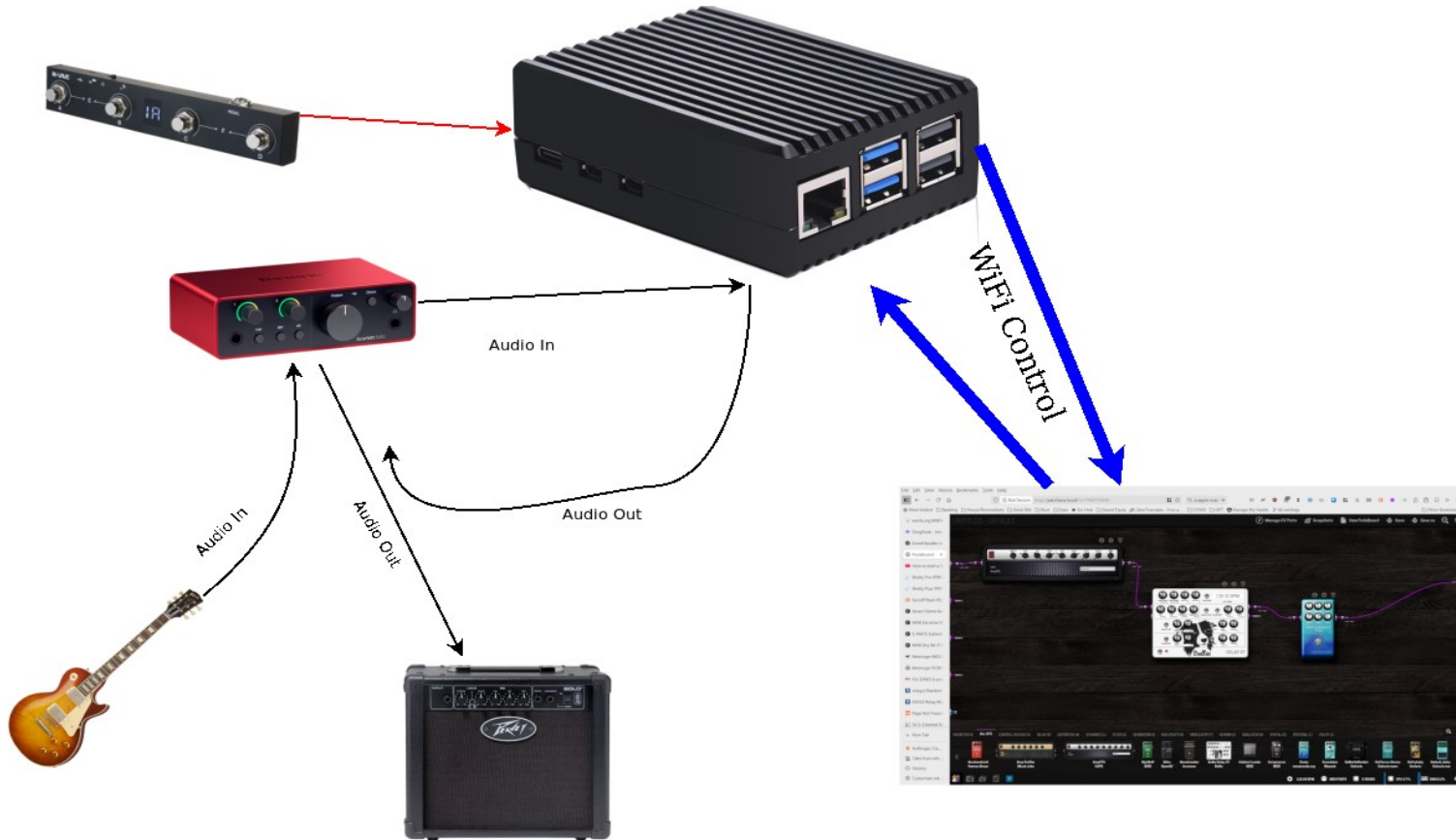
MIDI – Got to be

Pedals available – not common



Trigger on release?

Use Modep!?



500ms Latency
LV2 Only
Portability
Debian 12
Bit rot?

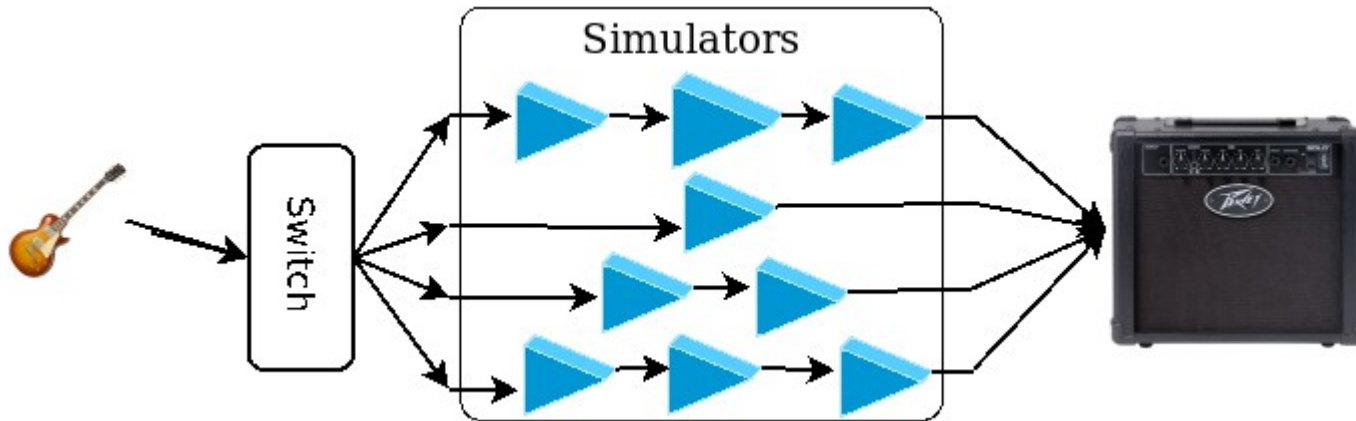
Mark I

Set up all the simulators in memory

Connect the “pedal-boards” with the Jack audio pipes between them

Do not set them up with I/O to the hardware devices

Write a MIDI controller to change Jack audio pipes to enable I/O



Digression iii: Modep/Modhost

Initialise: To UNIX socket -> mod-host

```
add add http://moddevices.com/plugins/mod-devel/BigMuffPi 7
param_set 7 Tone 0.500000
param_set 7 Level 0.500000
param_set 7 :bypass 0
param_set 7 Sustain 0.500000
```

Activate: Jack API or Shell

```
jack_connect effect_7:Out1 system:playback_1
jack_connect system:capture_1 effect_7:In
```

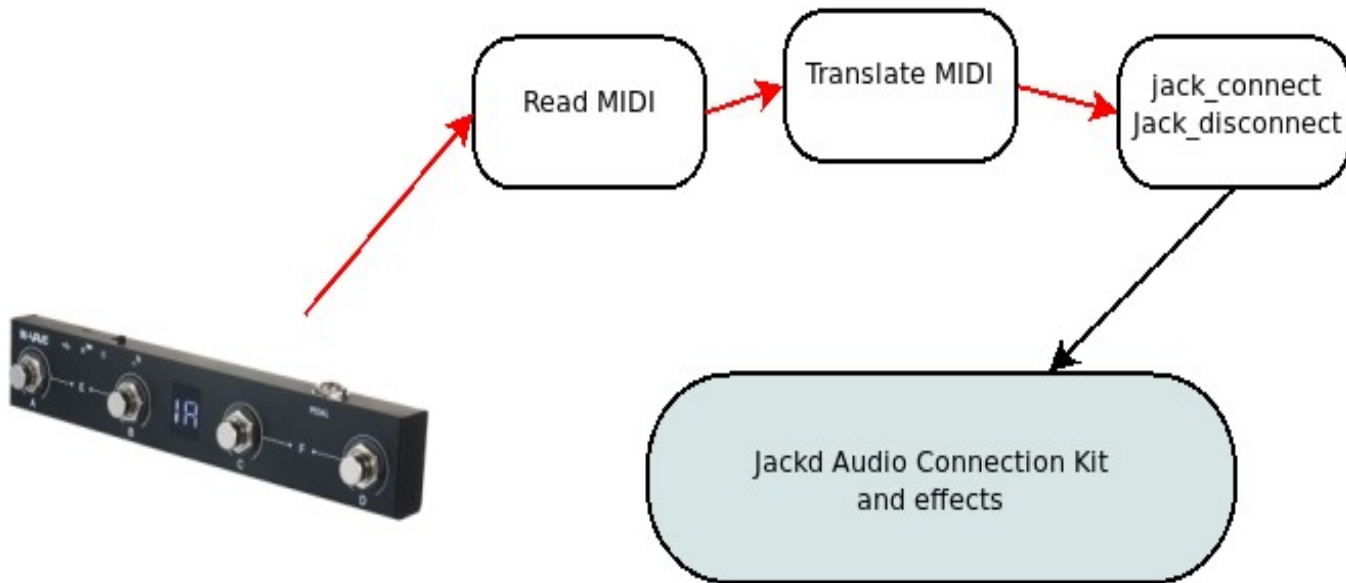
Modify: Unix socket -> mod-host

```
param_set 7 Tone 0.75
```



MIDI Driver

- * Read
 - * Translate
 - * Act
- ...much hand waving

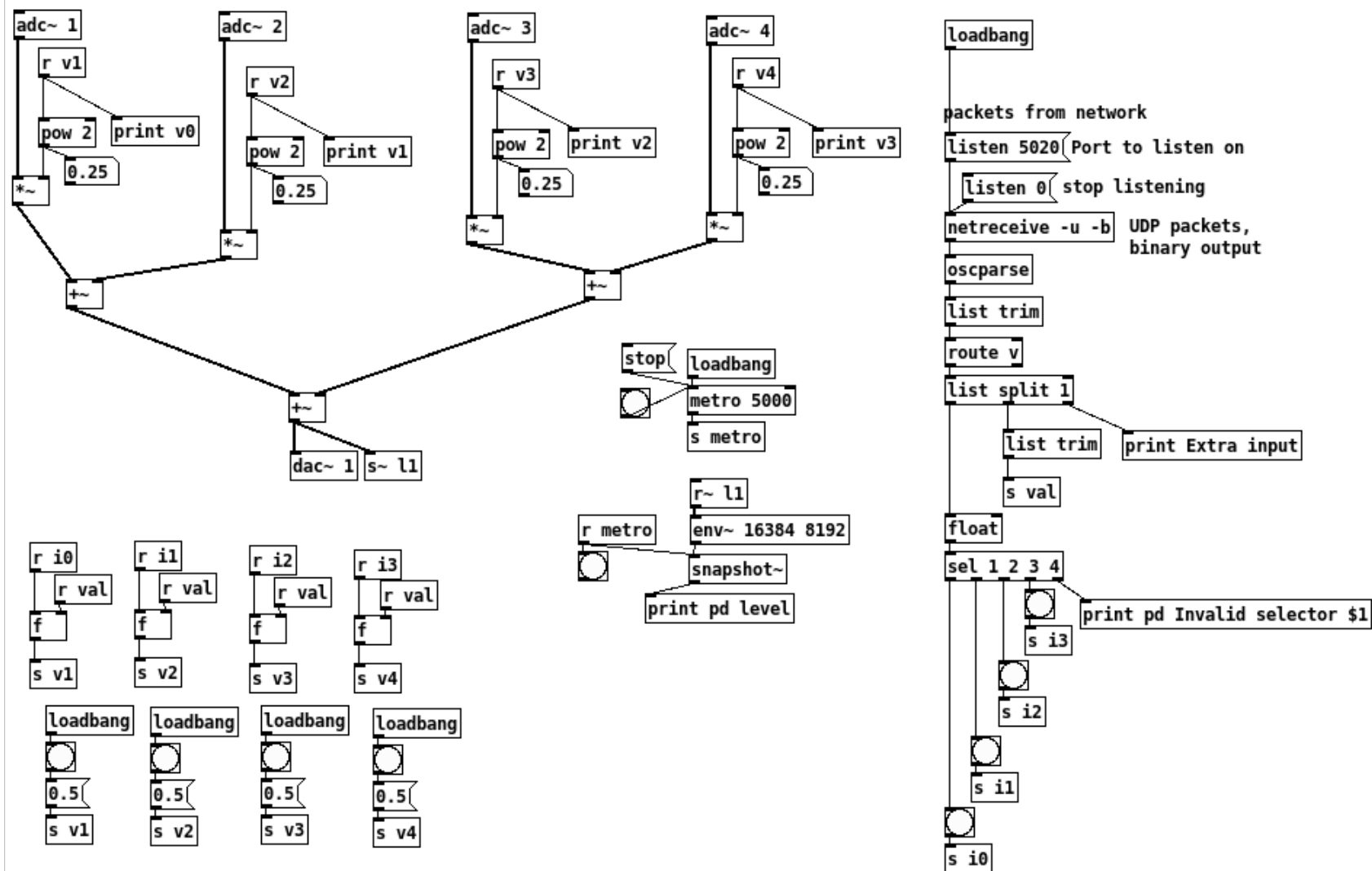


Not Finished Yet

- * Adjusting settings interrupts creative process
- * Most important setting is volume
- * Difficult to estimate subjective and relative volume of electric guitar on its own - as opposed to “full noise”
- * Use pd (Pure Data) to make simple mixer to control volume

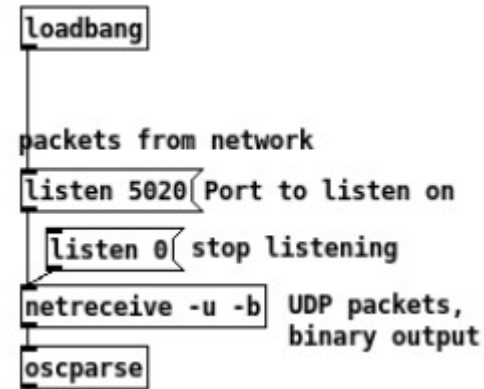


Patch to act as four to one channel mixer. The volume for each channel is set using OSC /v/<channel>/<volume>. The volume must be between 0..1 but is not checked.



Control Mixer With open Sound Control

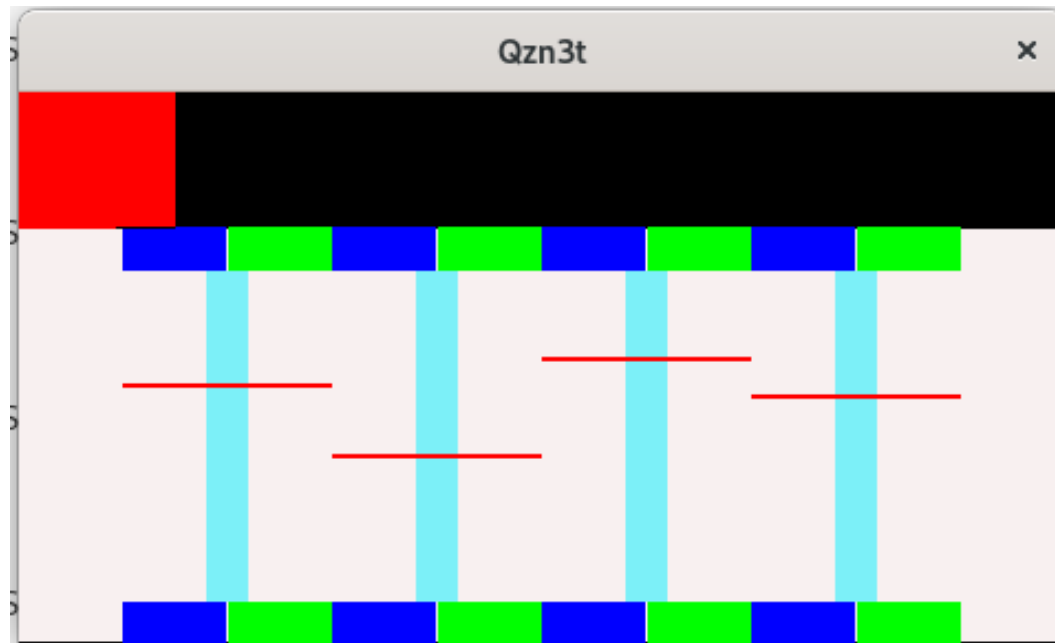
- * Pure Data listens on a port (5020)
 - * UDP packets
 - * `/v/2/0.8`
 - * Set effect 2 volume to 0.8
 - * Easy: Pure data has primitives to parse
 - * Rust crate `rosc` for encoding and transmitting
- Possibly easiest part of whole project



Control: UI

- * WIMP interface unsuitable
- * Keyboard on its own problematic
 - * moving parts
 - * super vulnerable to spills
- * Web-app c/f Modep?
 - * Yawn
 - * Require WiFi to operate
 - * Separate piece of equipment

Pi Touch Screen and Case



Description: Package includes --- an ABS case, 3.5 inch TFT LCD Display and a Pi 5 Active Cooler for Raspberry Pi 5. NOTE: Raspberry Pi Board not included. 

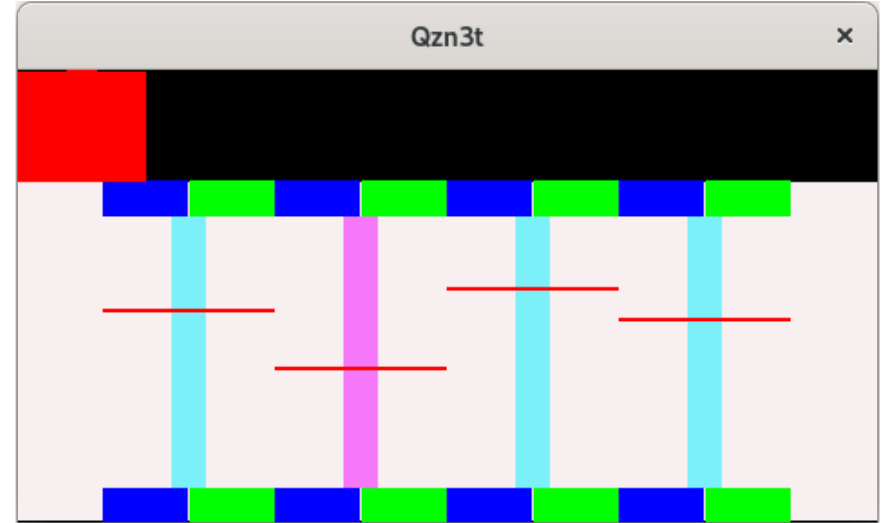
Regrets



State: Display, store...

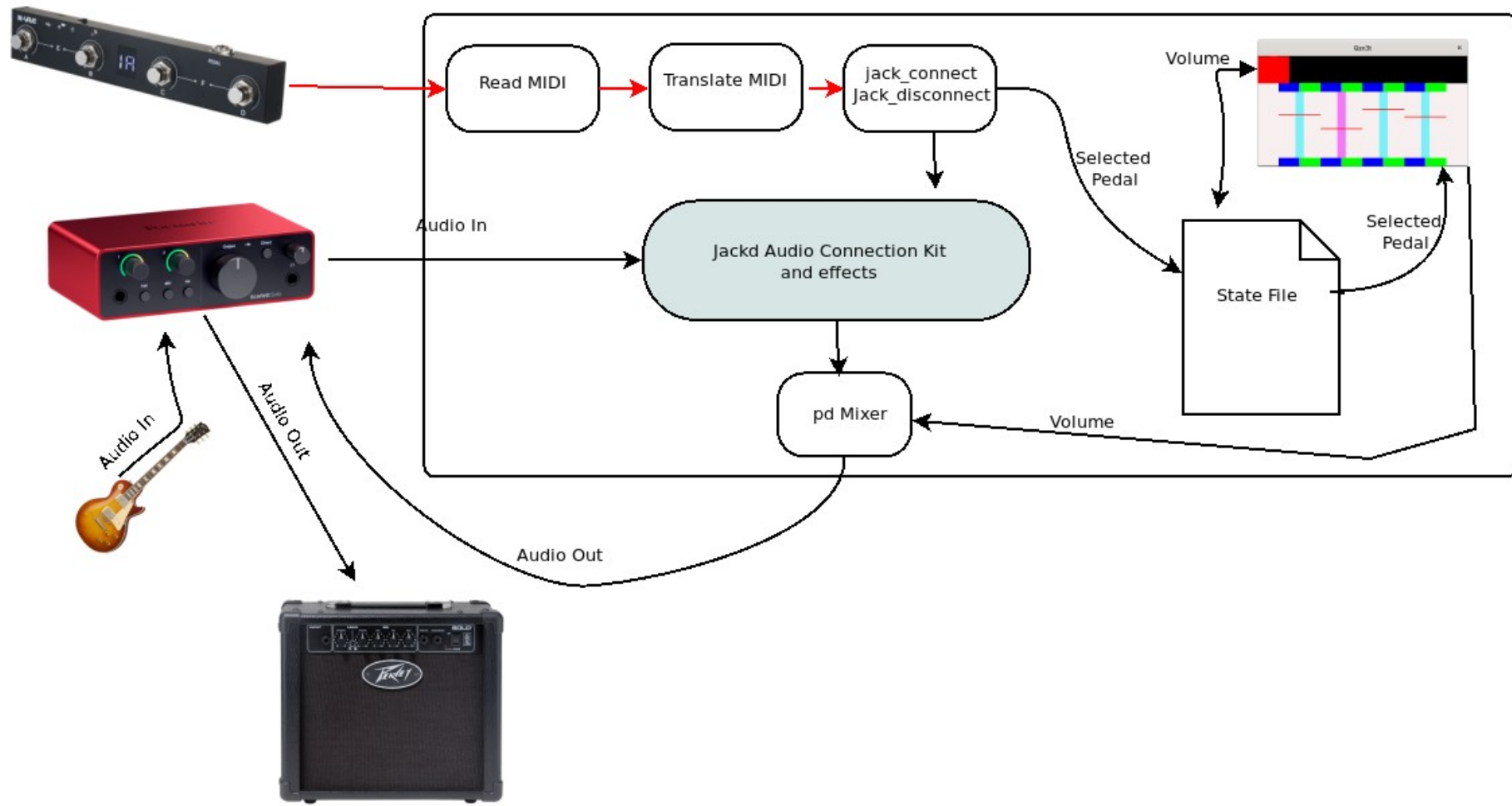
Added complications

Not enough to set volume, must be stored (and displayed) between rehearsal and performance



Take opportunity to display selected pedal

Finished?



Conclusion

- * Modep: Fabulous GUI.

- * Must have been huge effort over many years

```
$ find . -type f -name "*.c" -or -name "*.cpp" -or -name "*.py" -exec wc -l {} \; |  
perl -e '$t=0;while(<>){/^\d+/ or next; $t+=$_;}print("$t\n");'
```

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- * Useless live: That is what is it is all about (for me)

Complications

- * Basic switching simple
- * Interactions with pedal fraught (inconsistent MIDI)
- * Adjusting volume (relative volumes)
 - * Implies GUI
 - * Storing state

Next...

- Bugfixes
- Tuner
- Mixer “reset”
- Banks, and bank switching
 - Hard to get reliable for performance
- Web-app front end to meta configuration (Go)

The END?!?!

